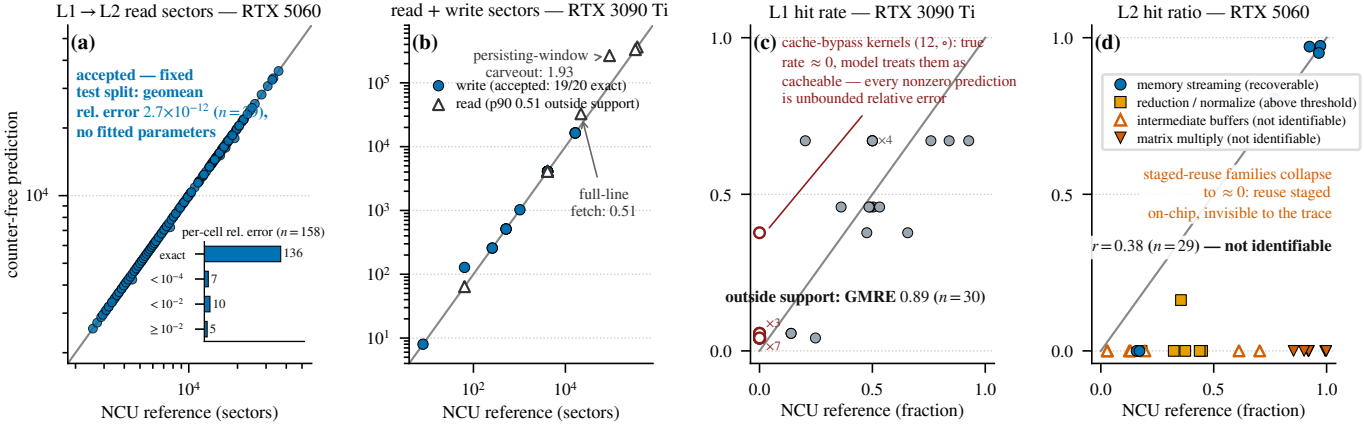
L1 → L2 read sectors — RTX 5060
(a)
accepted — fixed
test split: geomean
rel. error 2.7×10⁻¹² (n = 158),
no fitted parameters
per-cell rel. error (n = 158)
exact
136
< 10⁻⁴
7
< 10⁻²
10
≥ 10⁻²
5
counter-free prediction
NCU reference (sectors)
read + write sectors — RTX 3090 Ti
(b)
persisting-window > carveout: 1.93
write (accepted: 19/20 exact)
read (p90 0.51 outside support)
full-line fetch: 0.51
NCU reference (sectors)
L1 hit rate — RTX 3090 Ti
(c)
cache-bypass kernels (12, ∘): true rate ≈ 0, model treats them as cacheable — every nonzero prediction is unbounded relative error
×4
×3
×7
outside support: GMRE 0.89 (n = 30)
NCU reference (fraction)
L2 hit ratio — RTX 5060
(d)
memory streaming (recoverable)
reduction / normalize (above threshold)
intermediate buffers (not identifiable)
matrix multiply (not identifiable)
staged-reuse families collapse to ≈ 0: reuse staged on-chip, invisible to the trace
r = 0.38 (n = 29) — not identifiable
NCU reference (fraction)